

Plurals under quantification: disentangling readings and gradient effects

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Abstract We investigate the interpretation of plural expressions, starting from the puzzle of the logical gap between their interpretations in affirmative and negative sentences. While bare plurals (e.g. “books”) typically receive an *at least two* reading in upward-entailing environments (e.g. “The box contains books”), they yield an *at least one* reading in downward-entailing contexts (e.g. “The box doesn’t contain books”). Existing bivalent and trivalent theoretical approaches differ in their predictions about truth conditions of plural indefinites under the scope of a universal quantifier (e.g. “Every box contains books”) in mixed scenarios (some boxes contain one book, others contain several).

Through experimental studies in English, we uncover gradient truth-value judgments in such mixed scenarios, which challenge the categorical predictions of all available theories. Our results raise key questions about the empirical identification of accessible readings and we put forward a methodology to disentangle readings from gradient effects.

Keywords: Plurality, implicature, readings

1 Introduction

Plural nouns across many languages give rise to a well-known puzzle: their interpretation seems to be different in affirmative and negative sentences. This is illustrated in (1) for English bare plurals (Sauerland et al. 2005, Spector 2007, Zweig 2009):

- (1) a. The box contains books.
- b. The box doesn’t contain books.

In (1a), the noun “books” typically gives rise to a *multiplicity inference*, i.e. the sentence is understood to convey that the box contains at least two books. However,

(1b) is not understood to convey the logical negation of this proposition, since if it were, the sentence would be judged true as soon as the box does not contain at least two books, and thus, in particular, when the box contains exactly one book. In fact, the sentence is true if and only if the box does not contain any books at all, suggesting that the bare plural is now interpreted as meaning *at least one book*. This creates a kind of ‘logical gap’: the perceived meaning of (1b) is not the classical logical negation of that of (1a), and there are situations where neither (1a) nor its negation (1b) is perceived to be clearly true.

More generally, the *at least one* reading arises in downward-entailing (DE) environments, i.e. environments that reverse the direction of logical entailment¹ (as already noted in Sauerland et al. 2005, Spector 2007, Zweig 2009). By contrast, “books” generally receives an *at least two* reading in upward-entailing (UE) environments, i.e. environments that preserve the direction of logical entailment, such as simple affirmative sentences like (1a). Several theoretical accounts have been proposed to capture this generalization:

- (i) **Implicature approaches:** these treat the multiplicity inference as a scalar implicature. On this view, bare plurals mean *at least one*, and the stronger *at least two* reading arises through strengthening via the $\langle \text{PL}, \text{SG} \rangle$ scale (Sauerland et al. 2005, Spector 2007, Zweig 2009). We focus on two main versions: the Higher-Order Implicatures (henceforth, HOI) approach from Spector (2007) and the implicature approach based on Zweig (2008) and Zweig (2009), with additions from Ivlieva (2020).
- (ii) **Trivalent approaches:** the homogeneity-based trivalent approach encodes the logical gap directly in the truth conditions via a third, ‘undefined’ value² (Križ 2017, building on Križ 2016); the Presuppositional Exhaustification approach is a hybrid account in which the multiplicity inference is derived as a scalar implicature, but scalar implicatures are treated as being in some sense presuppositions, within a trivalent semantics approach (Ahn et al. 2020, Bassi et al. 2021). Ultimately, these two approaches make the same predictions for the sentences under study here.³

1 Formally, f is downward-entailing if whenever $P \models Q$, then $f(Q) \models f(P)$.

2 If one takes homogeneity to be a presupposition, then the homogeneity-based approach could be seen as a kind of presuppositional approach. The precise status of homogeneity lies beyond the scope of this paper.

3 Farkas & de Swart (2010) proposed an approach where plural nouns are ambiguous between an inclusive ‘at least one’ interpretation and an exclusive ‘at least two’ interpretation, and disambiguation is governed by a general preference for stronger meanings. As discussed in Spector (2007) and Ivlieva (2020), and as acknowledged by Farkas & de Swart (2010) themselves, the ambiguity approach cannot account for the interpretation of bare plurals and plural indefinites in non-monotonic contexts.

A key question that bears on how to distinguish these approaches concerns the accessibility of different readings for more complex sentences, in particular sentences in which a bare plural is interpreted in the scope of a universal quantifier, such as (2):

(2) Each box contains books.

- (i) **Literal** reading: each box contains at least one book.
Scenarios that satisfy *only* the literal reading will henceforth be called *uniformly-singular* scenarios.
- (ii) **Weak** reading: each box contains at least one book and it is not the case that each box contains exactly one book.
Scenarios that satisfy *only* the literal and the weak readings will be called *mixed* scenarios.
- (iii) **Strong** reading: each box contains several books.
Scenarios that satisfy the strong reading (and therefore satisfy all three readings) will be called *uniformly-plural* scenarios.

These readings are ordered by logical strength: strong > weak > literal.

As we shall see, existing theories differ with respect to whether they predict the weak reading to exist as a distinct reading. The main goal of this paper is to help adjudicate between theories by experimentally investigating the availability of these three readings. Importantly, in our experiments, we control for a crucial confounding factor that previous work has overlooked: truth-value judgments in experimental tasks – especially (but not only) when participants are given more than two answer options (e.g. Likert scale or continuous scale) – may reflect, in addition to truth-value evaluation, *gradient effects*, whereby different instantiations of the very same readings might systematically receive different ratings. A secondary contribution of this paper is methodological, as it proposes a way to disentangle such non-purely truth-conditional gradient effects from the effects of the distinct *readings* which theories posit for a given sentence type.

That being said, if the preference for stronger meanings is replaced with a principle according to which a sentence with a bare plural counts as true if it is true under both the exclusive and inclusive interpretations, this problem is resolved, and the resulting system is then extremely similar to a trivalent approach. See, e.g. [Spector \(2025\)](#), which argues that trivalent Presuppositional Exhaustification approach can be reconstructed in terms of ambiguity between parses with or without an exhaustivity operator.

2 Theoretical background

We now turn to a brief overview of the theoretical intuitions and mechanisms proposed by each approach to derive the predicted readings for a sentence like (2).

2.1 Implicature approaches

Implicature approaches are based on strengthening with respect to alternatives, and predictions differ depending on what alternatives are taken into account. The informal reasoning from the HOI approach is as follows, for non-quantified sentences:

- (3) a. I read books.
b. I read a book.

The literal meaning of (3a) is *I read at least one book*.⁴ In order for its meaning to get strengthened to *I read more than one book*, (3a) needs to have a competitor whose meaning is *I read exactly one book*. “I read exactly one book” is neither a plausible alternative to (3a) (for reasons of syntactic complexity), nor the literal meaning of (3a)’s actual alternative, (3b). However, *I read exactly one book* is the *strengthened* meaning of (3b). Thus, (3a) is strengthened relative to the strengthened meaning of (3b). Because two strengthenings take place, the pragmatic meaning is said to be derived through *higher-order* implicature.

Let us now turn to quantified sentences.

- (5) a. Each box contains books.
b. Each box contains a book.

The literal meaning of (5a) is *each box contains at least one book*. (5b) is taken to be its competitor. When only “a book” contributes to the alternatives of (5b), then (5b) gets the strengthened meaning *each box contains at least one book and it is not the case that each box contains several*. (5a), in turn, is strengthened via the negation of (5b)’s strengthened meaning, which yields the strong reading: *each box contains several books*. According to the HOI approach, when both “every”

⁴ The implementation is based on a mereological theory of plurality (following Link et al. 1983, Link 1990, Landman 2000, a.o.) where plural nominal morphology is a function that takes as input a set P of atomic individuals and returns its closure under sum, notated $^\oplus P$. For example:

$$(4) \quad \begin{aligned} \llbracket \text{book} \rrbracket &= \{a, b, c\} \\ \llbracket \text{books} \rrbracket &= \llbracket \text{book-PL} \rrbracket = {}^\oplus \llbracket \text{book} \rrbracket = \{a, b, c, a \oplus b, b \oplus c, a \oplus c, a \oplus b \oplus c\} \end{aligned}$$

In this sense, a bare plural is semantically number-neutral and has the literal meaning *at least one*.

and “a book” contribute to the alternatives of (5b) (i.e. when the derivation takes into account both an alternative to “every”, namely “some”, and an alternative to “a book”, namely “several books”), then (5b) gets the strengthened meaning *each box contains exactly one book*. The strengthening of (5a), in this case, yields the weak reading: *each box contains at least one book and it is not the case that they all contain exactly one*.

Another implicature approach was proposed by [Zweig \(2009\)](#) and later reformulated by [Ivlieva \(2020\)](#) using an exhaustivity operator. This approach makes slightly different predictions, as it yields the strong reading whenever the exhaustivity operator is applied. Although the strong reading is not the only candidate meaning generated by the system, Zweig’s approach selects it via the Strong Candidate Principle, which gives preference to the strongest among the candidate meanings. Since the operator can be either applied or not applied at each scope site, the literal reading is also predicted and corresponds to the configuration in which the operator is never applied.

2.2 Trivalent approaches

We present two trivalent approaches, one proposed by [Križ \(2017\)](#) and another by [Ahn et al. \(2020\)](#). Both employ the Strong Kleene framework to derive the semantics of quantifiers over atoms applied to trivalent predicates. The Strong Kleene semantics for the universal quantifier, with a bivalent restrictor P and a trivalent scope Q , is defined as:

$$(6) \quad \llbracket \text{Each } P \text{ is } Q \rrbracket = \begin{cases} 1 & \text{if } \forall x. \llbracket P \rrbracket(x) = 1 \rightarrow \llbracket Q \rrbracket(x) = 1, \\ 0 & \text{if } \exists x. \llbracket P \rrbracket(x) = 1 \wedge \llbracket Q \rrbracket(x) = 0, \\ \# & \text{otherwise.} \end{cases}$$

In [Križ \(2017\)](#), it is proposed that the trivalent approach to homogeneity (cf. [Križ 2016](#), with roots in [Löbner 2000](#)) can be extended to plural indefinites and bare plurals. In this approach, the undefined value captures the logical gap in the truth conditions of the non-quantified sentence:

$$(7) \quad \llbracket \text{Box } b \text{ contains books} \rrbracket = \begin{cases} 1 & \text{if } b \text{ contains several books,} \\ 0 & \text{if } b \text{ contains no books,} \\ \# & \text{if } b \text{ contains exactly one book.} \end{cases}$$

At the sentence level, this yields the following trivalent denotation for a sentence like (2):

$$(8) \quad \llbracket \text{Each box } \lambda_b [b \text{ contains books}] \rrbracket = \begin{cases} 1 & \text{if each box contains several books,} \\ 0 & \text{if there is at least one empty box,} \\ \# & \text{otherwise.} \end{cases}$$

Ahn et al.’s (2020) approach (which builds on Bassi et al. 2021) relies on *Presuppositional Exhaustification* and motivates trivalence differently from Križ (2017). Here, the undefined value is assigned when the relevant sentence is true under its ‘literal’ meaning but not under its strengthened meaning. Formally, the Presuppositional Exhaustification operator is defined as:⁵

$$(9) \quad \llbracket \mathbf{PEX}(p) \rrbracket = \lambda x. \begin{cases} 1 & \text{if } \llbracket p \rrbracket(x) = 1 \text{ and } \bigwedge_{\substack{q \in \text{ALT}^*(p) \\ p \not\leq q}} \llbracket \neg q \rrbracket(x) = 1, \\ 0 & \text{if } \llbracket p \rrbracket(x) = 0, \\ \# & \text{otherwise.} \end{cases}$$

As argued in Ahn et al. (2020), **PEX** applies locally to the plural NP, whose alternative is the corresponding singular NP.

$$(10) \quad \llbracket \mathbf{PEX}(\text{books}) \rrbracket = \lambda x. \begin{cases} 1 & \text{if } x \text{ is a plurality of books,} \\ 0 & \text{if } x \text{ does not contain any books,} \\ \# & \text{otherwise, e.g. if } x \text{ is an atomic book.} \end{cases}$$

The approach posits that application of operator is obligatory. This returns the same truth conditions as the homogeneity-based approach at the sentence level:

$$(11) \quad \llbracket \text{Each box } \lambda_b [b \text{ contains } \mathbf{PEX}(\text{books})] \rrbracket = \begin{cases} 1 & \text{if each box contains several books,} \\ 0 & \text{if at least one box is empty,} \\ \# & \text{otherwise.} \end{cases}$$

To sum up, in both trivalent approaches, the undefined value collapses uniformly-singular situations and mixed situations. Therefore, these approaches assign no special status to the weak reading. As for the literal reading, it corresponds to all situations in which the relevant sentence is not false (true or undefined, i.e. *tolerantly true* in Cobreros et al.’s 2012 approach). Under the trivalent approach, it is expected

⁵ In (9), in the case of plural indefinites, $\text{ALT}^*(p)$ contains a single element, which is $\lambda x. p(x) \wedge \text{ATOM}(x)$.

that the sentence should be rated higher in situations where it is not false but not strictly true (i.e. situations in which it is undefined) than in situations where it is plainly false (cf. [Križ & Chemla 2015](#)), hence we should be able to detect a contrast between the two types of scenarios in an experimental setting.⁶

2.3 Summary

We summarize below the readings predicted by each approach:

HOI approach	{ literal, weak, strong }
Zweig(+Ivlieva)’s approach	{ literal, strong }
Trivalent approaches	{ literal, strong }

3 Previous experimental studies and gradient effects

Few experimental studies have tried to specifically detect the weak reading, and the few that did report conflicting results.

[Farkas & de Swart \(2010\)](#), who argue for an approach based on ambiguity (cf. footnote 3), report the results of a small-scale study in which 24 participants were asked to judge whether the sentence “Each sportsman is wearing gloves” is true in mixed scenarios, where some sportsmen wear two gloves, and others only one. To ensure that wearing only one glove was not unnatural, some of the relevant sports typically required only one glove (e.g. baseball). They report that 23 participants judged the sentence false. In contrast, the ‘control’ sentence “All children were sitting on small chairs” was judged true by 22 participants in a situation where each child is seated on exactly one chair – a judgment that is attributed to a cumulative interpretation of the sentence. They argue on this basis that the weak reading is not available. While these findings are suggestive, we note that (i) even if different sports were used, the use of *gloves*, which typically go by two, is still likely to favor the strong reading, as opposed to other lexical items, and (ii) the control sentence

⁶ [Križ \(2016\)](#) describes a way in which an undefined sentence may still be considered *true enough* relative to a given Question Under Discussion (QUD). A QUD partitions the set of possible worlds into equivalence classes, or cells, corresponding to the distinctions that matter in the conversational context. A sentence that is undefined in the actual world can nevertheless be used felicitously if, within the QUD-cell that contains the actual world, there is a world where the sentence is true and there is no world where the sentence is false. That cell may include worlds where the sentence is undefined. In such a case, the worlds where the sentence is true are indistinguishable, from the QUD’s perspective, from the actual world (where the sentence may be undefined). It is therefore acceptable, given the QUD, to utter the sentence *as if* it were true.

and the target sentence do not form a minimal pair (they would if the control item had been “All sportsmen are wearing gloves”, in the same scenario).

Two other previous studies contain data relevant to our inquiry: [Stateva et al. \(2016\)](#) and [Jiang & Sudo \(2023\)](#). While [Stateva et al. \(2016\)](#) investigated the existence of the weak reading, [Jiang & Sudo \(2023\)](#) did not target this question but reported results from separate experiments that are potentially informative for our purposes. Both collected semantic judgments on a continuous (Likert) scale, presenting participants with uniformly-singular scenarios, mixed scenarios and uniformly-plural scenarios. These designs are informative for determining which readings are actually accessed: if three distinct readings exist, then mixed scenarios, which make true two of the three readings, should be rated higher than uniformly-singular scenarios, which only make true the literal reading, but lower than uniformly-plural scenarios, which make all three readings true (following the same logic as in [Cremers & Chemla 2016](#), [Stateva et al. 2016](#), [Chemla & Spector 2011](#), [Denić & Sudo 2022](#), among others). All theoretical accounts agree that a uniformly-plural scenario (Figure 2) should receive higher ratings than a uniformly-singular scenario (Figure 3). If mixed scenarios (Figure 4) are rated in-between, this can be taken as evidence that the weak reading is genuinely available.

The experiments reported in [Jiang & Sudo \(2023\)](#) were designed to address a different research question (the role of context, and specifically questions under discussion, in licensing readings without a multiplicity inference) but included mean ratings for bare plurals under universal quantification (e.g. “every boy opened presents”) in uniformly-singular scenarios (every boy opened one out of two presents) and mixed scenarios (half of the boys opened one out of two presents, while the other half opened both). Participants were asked to use a 5-point Likert scale (ranging from “completely false” to “completely true”) to rate each sentence against a scenario. Uniformly-singular and mixed scenarios were tested in separate experiments. In each case, a distinct context was provided in the form of a family rule that required either keeping presents closed (“You must keep each present you received closed!”) or opening them (“You must open each present you received!”). Each context was intended to favor or disfavor a weak reading of the bare plural, which the authors refer to as the lax and strict contexts, respectively. Despite these scenarios being tested in two separate experiments, we can still compare their mean ratings. The comparison between the uniformly-singular scenarios of their Experiment 1 with the mixed scenarios of their Experiment 2 yields 4.68 vs 4.3 in contexts thought to favor a literal reading and 3.82 vs 4.04 in contexts thought to favor the strong reading. This provides no clear evidence for the availability of the weak reading, since the direction of change between the two scenarios is inconsistent across contexts, in a way that no current theory lets us expect.

Stateva et al. (2016) presents two truth-value judgment experiments on plural indefinites under universal quantification in Italian (“ogni coccinella ha alcuni/dei puntini”, *every ladybug has (some) dots*). The experiments differed in the plural expression used (“alcuni puntini” vs. “dei puntini”) but shared the same set of conditions, in which the proportion of single-dotted ladybugs was 0, 1, 3 or 5 out of 5, with all ladybugs having at least one dot. Participants were asked to use a 7-point Likert scale (ranging from “definitely inappropriate” to “definitely appropriate”) to rate each sentence against a scenario.

The results are reproduced in Figure 1. Note that the x -axis is the proportion of single-dotted ladybugs. In both graphs, the condition ‘0-of-5’ is a uniformly-plural scenario, ‘1-of-5’ and ‘3-of-5’ are mixed scenarios, and ‘5-of-5’ is the uniformly-singular scenario.

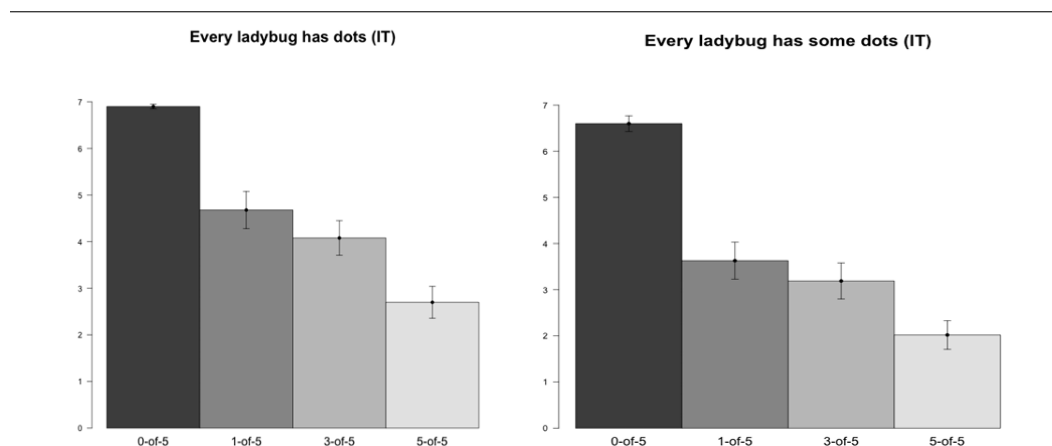


Figure 1: Results from Stateva et al. (2016). Mean acceptability ratings for sentences of the form “ogni coccinella ha dei puntini” (left, *every ladybug has dots*) and “ogni coccinella ha alcuni puntini” (right, *every ladybug has some dots*). X-axis: proportion of single-dotted ladybugs.

In both experiments, significant differences were found between every pair of conditions except between the two middle conditions (‘1-of-5’ and ‘3-of-5’), which correspond to mixed scenarios. In our terms, the analysis therefore supported the existence of both the weak and the strong readings.

Looking at the mean results, two consistent patterns also emerge:

- (i) a gradual increase in mean ratings as the number of multi-dotted ladybugs increased, including within the mixed-scenario conditions, i.e. ‘1-of-5’ and ‘3-of-5’ (despite the lack of a significant difference);
- (ii) a sharp rise in acceptability precisely at the point where the strong reading becomes true (i.e. the uniformly-plural scenario).

Theoretically, (ii) suggests that the strong reading is robustly available and highly salient. In light of (i), an alternative account of the data suggests itself: maybe the only two readings are the literal and strong readings, and the intermediate rate for the mixed scenarios reflects an independent gradient effect. That is, among conditions that do not make the strong reading true, those that are *more similar* to a uniformly-plural scenario, i.e. where there are more multi-dotted ladybugs, receive a higher rating. Note, in particular, that the absence of a statistically significant difference between the ‘1-of-5’ and ‘3-of-5’ conditions does not, on its own, demonstrate that no effect exists. Because the original studies did not control for gradience, it remains possible that all of the observed contrasts were driven solely by gradience, including the contrast between the uniformly-singular and the mixed scenarios (‘5-of-5’ vs. ‘3-of-5’). What we would need to establish in order to rule out such an interpretation is that the difference between the uniformly-singular condition (‘5-of-5’) and the closest mixed condition (‘3-of-5’) is stronger than the one between conditions that are both mixed (‘1-of-5’ vs. ‘3-of-5’) – but the fact that one comparison is significant while the other is not does not demonstrate this.⁷

The plausibility of such additional gradient effects can be made salient by means of the following thought experiment, which will serve as a basis for our actual experiments reported in Sections 4, 5, 6 and 7. Imagine participants have to rate the sentence “each box contains books” against pictures as in Figures 2 (uniformly-plural scenario), 3 (uniformly-singular scenario) and 4 (mixed scenarios). Intuitively we expect the highest ratings for the uniformly-plural scenario (Fig. 2) and the lowest ones for the uniformly-singular scenario (Fig. 3).

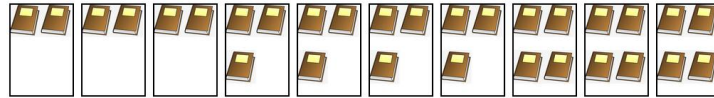


Figure 2: Example of a uniformly-plural scenario for “each box contains books”.

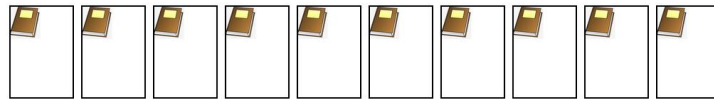


Figure 3: Uniformly-singular scenario for “each box contains books”.

⁷ Cf. Gelman & Stern (2006), “The difference between ‘significant’ and ‘not significant’ is not itself statistically significant”.

Regarding mixed scenarios, it seems intuitive that they should not all be assigned the same score; rather, ratings might increase with the number of ‘strong verifiers’ (individuals in the quantification domain that satisfy a multiplicity inference) within the scenario. In other words, even though these scenarios make exactly the same readings true (the literal and weak readings), participants would behave as if they were ordered along a scale which appears to correspond to similarity to the uniformly-plural scenario.

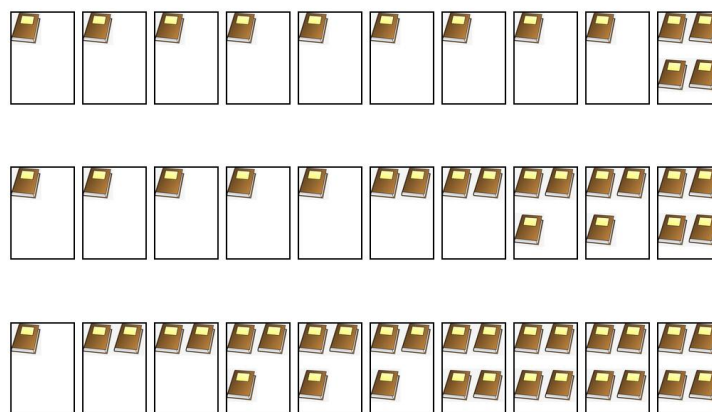


Figure 4: Examples of mixed scenarios for “each box contains books”.

Now, as we saw, there is some evidence for such gradient effects in [Stateva et al.’s \(2016\)](#) results, but also in related experimental work on scalar implicatures ([Chemla & Spector 2011](#), [van Tiel 2014](#)). As we discussed, the potential presence of gradience has an unfortunate consequence: it can create the illusion of a weak reading, especially if only one type of mixed scenario is tested (say, one where half of the boxes are ‘strong verifiers’). That is, while an intermediate rating for mixed scenarios (higher than uniformly-singular scenarios but lower than uniformly-plural scenarios) might initially be taken as evidence in favor of the availability of the weak reading, it might turn out to simply reflect the fact that mixed scenarios, in comparison to uniformly-singular scenarios, are closer to uniformly-plural scenarios. Importantly, neither [Stateva et al. \(2016\)](#) nor [Jiang & Sudo \(2023\)](#) explicitly controlled for this confound, even though the data in [Stateva et al. \(2016\)](#) already suggested the presence of variation within mixed scenarios.

Detecting the weak reading is therefore particularly difficult in the presence of gradience, since we need to disentangle gradient effects from differences in readings: quantitative variation due to gradience may either obscure or be mistaken for qualitative differences between readings. The aim of this paper is to determine which readings are genuinely available while controlling for possible gradient effects.

To conclude, given the possibility of gradient effects, if we observe intermediate ratings for mixed scenarios, we need to adjudicate between two types of account:

- (i) accounts that predict only literal and strong readings, with intermediate ratings for mixed scenarios arising from an independent gradient effect, based on a notion of similarity to a uniformly-plural scenario;
- (ii) accounts that predict the availability of a weak reading on top of the literal and strong readings, without excluding an independent gradient effect in addition to these three readings.

Our experimental designs and statistical modeling will allow us to adjudicate between these two possibilities.

4 Comprehension study on bare plurals: continuous judgments⁸

4.1 Methods

The main novelties of our experimental design are the inclusion of literally false scenarios and the ability to capture gradience within classes of scenarios that make true exactly the same readings (specifically, scenarios where the sentence is false under all readings, and mixed scenarios). The latter was made possible by including more than two literally false conditions, and more than two mixed conditions. This was not the case in previous work (Stateva et al. 2016, Jiang & Sudo 2023), where experiments did have mixed conditions, but only had at most two.

Participants. We tested 200 adult participants recruited through Prolific (mean age 41.7; age range 19-77; 119 female). The sample size was based on power analyses from a pilot study. Participation was restricted to Prolific workers located in the UK or the US who reported English as their ‘first language’, ‘primary language’, and ‘earliest language in life’. They were paid £0.75 for their participation.

Procedure and stimuli. A trial consisted of a picture and a sentence. The picture consisted of four boxes containing between 0 and 4 geometric shapes, which were identical in all boxes (circles, triangles, or squares). The sentence followed a fixed structure: “Each box contains [bare plural]”, where the [bare plural] corresponds to the geometric shape present in the picture.

The experimental conditions are determined by two factors: (a) which of the three putative readings are made true by the picture, (b) the number of *strong verifiers*, namely the number of boxes that contain more than one shape. This gave rise to 9 conditions in total (cf. Figure 5), which can be described as follows:

⁸ All preregistration materials for our experiments are available anonymously in [this OSF project](#). A link to the original pre-data-collection registration will be added after review.

- **4 FALSE conditions:** no reading is true, i.e. at least one box is empty, while the non-empty ones, if any, are always strong verifiers. The number of strong verifiers ranges from 0 to 3, giving rise to 4 distinct conditions.
- **1 LITERAL condition:** only the literal reading is true, i.e. each box contains exactly one shape, and there is of course no strong verifier;
- **3 WEAK conditions:** the literal reading and the weak reading are true, but not the strong reading (each box contains at least one shape, at least one box contains more than one, and at least one box contains exactly one). The number of strong verifiers ranges from 1 to 3, giving rise to 3 distinct conditions.
- **1 STRONG condition:** all readings are true (each box contains at least two shapes, i.e. is a strong verifier).

Each trial was accompanied by the same instruction: “Use the cursor to indicate how well you think the sentence below describes the image.” The cursor moved on a continuous scale ranging from “bad description” (left extremity) to “good description” (right extremity). Participants’ responses were recorded as integers between 1 and 100, but the numerical rating was not visible to participants. Figure 6 shows an example of a trial.

Each participant completed 27 randomized trials, with each condition appearing three times, using three different geometric shapes in three different colors.

The final page included an attention check.

4.2 Results

Participants who completed the full experiment were excluded if:

- (i) they failed the attention check (2 participants);
- (ii) they did not rate every trial of the **STRONG-4** condition higher than every trial of the **FALSE-0** condition (10 participants).

We defined four predictors:

- c_{vrf} Number of strong verifiers.
- c_{lit} Binary variable indicating whether the condition supports the literal reading.
- c_{weak} Binary variable indicating whether the condition supports the weak reading.
- c_{str} Binary variable indicating whether the condition supports the strong reading.

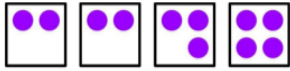
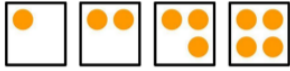
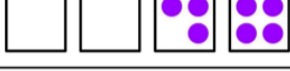
stimuli pictures (examples with circles)	label for condition	readings
	STRONG-4	true for STRONG+WEAK+LITERAL readings
	WEAK-3	true for WEAK+LITERAL readings
	WEAK-2	true for WEAK+LITERAL readings
	WEAK-1	true for WEAK+LITERAL readings
	LITERAL-0	true for LITERAL reading
	FALSE-3	false for LITERAL reading
	FALSE-2	false for LITERAL reading
	FALSE-1	false for LITERAL reading
	FALSE-0	false for LITERAL reading

Figure 5: Stimuli for bare plurals, with corresponding readings.

Results are shown in Figure 7, with the mean ratings and histograms of the ratings for each condition.

We subset the literally true conditions (**LITERAL**, **WEAK**, **STRONG**) and fit a linear mixed-effects model predicting responses as a function of c_{vrf} , with random intercepts and slopes by participant:

$$\text{response} \sim c_{\text{vrf}} + (1 + c_{\text{vrf}} \mid \text{participant})$$

The random slope was removed due to failure of convergence.

We compared this model to a null model containing only the intercept using a likelihood ratio test (LRT). As expected, the effect of c_{vrf} was positive ($b = 4.93$,

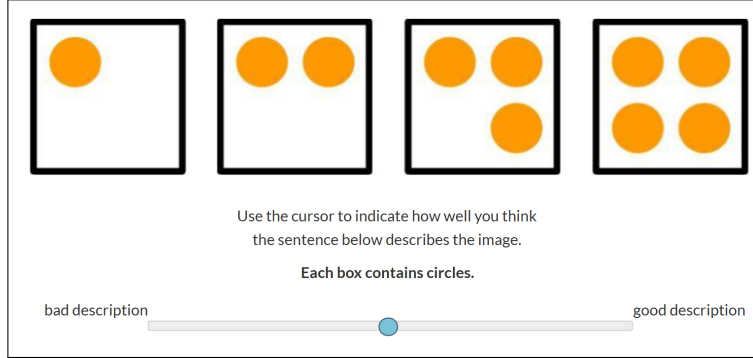


Figure 6: Example of a trial in the comprehension study with bare plurals.

$SE = 0.24$, $t = 20.64$) and the model fit on literally true conditions (c_{vrf} as predictor) significantly outperformed the null model ($\chi^2(1) = 395.01$, $p < 10^{-15}$).⁹

We also performed the LRT on the **WEAK** conditions, to test for gradient effects within conditions that all make true the same readings. Crucially, a gradient effect was also found within **WEAK** conditions alone ($\chi^2(1) = 18.75$, $p < 10^{-4}$) (this time with the random slope, which did not prevent model convergence).

As an exploratory analysis (not preregistered), we conducted model comparisons across all 9 conditions using the Bayesian information criterion (BIC) and the Akaike information criterion (AIC) to identify the best-fitting combination of predictors, among the $2^4 = 16$ possible combinations. Our primary claims will be based on the BIC criterion, which gives a stronger penalty for model complexity.

The best-fitting model across all 9 conditions was:

$$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$$

Table 1 reports statistical details of the best model.

The second best-fitting model was:

$$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$$

with $\Delta BIC = 8$ between the two best models. This difference is considered strong evidence in favor of the better-ranked model, following standard guidelines (e.g. Raftery 1995).

The full model rankings using BIC and AIC are presented in Table 2. Importantly, both criteria yielded the same order of ranking.

⁹ The p -values reported have been corrected for multiple comparisons using the Holm-Bonferroni method.

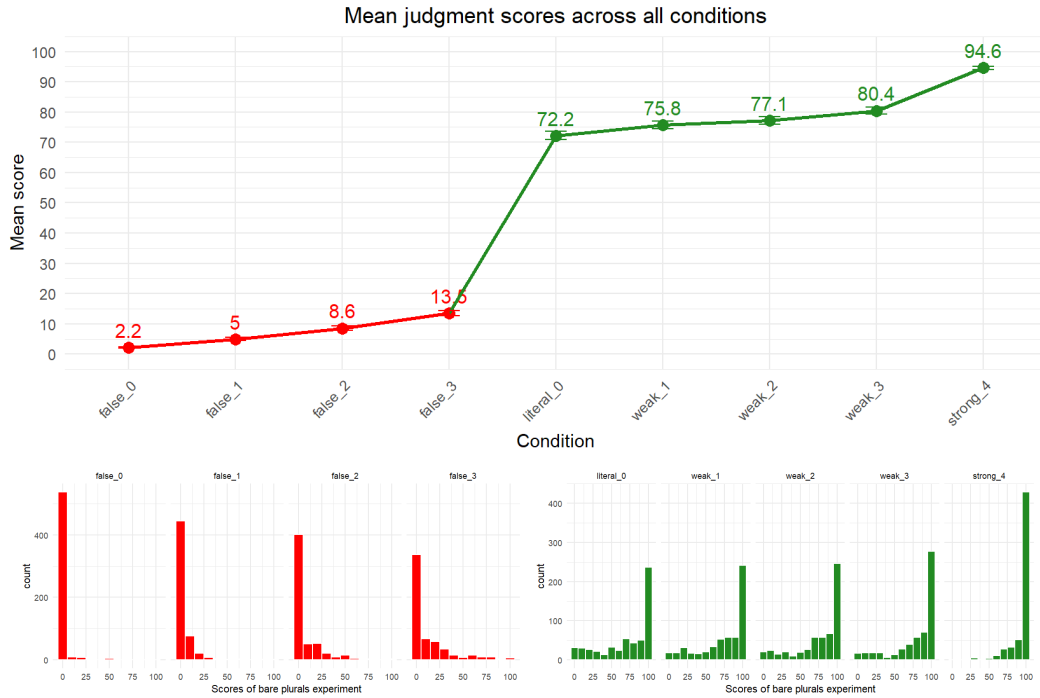


Figure 7: Results of the comprehension task with bare plurals.

4.3 Discussion

Our results showed gradient effects driven by c_{vrf} in both true and false conditions and, crucially, also within **FALSE** and within **WEAK** conditions. Gradience across conditions of the same reading is evidence for the status of gradience as a factor of its own (confirmed by the model comparisons), and not as a by-product of increased judgment scores when one more reading is made true.

We observed a qualitative shift in judgments between the **WEAK** conditions and the **STRONG-4** condition, but only a quantitative increase (driven by c_{vrf}) from **LITERAL-0** to **WEAK-1**. The post-hoc analyses comparing models in terms of BIC confirm this qualitative description, since the best model is the one that includes as predictors the number of strong verifiers, the truth-value of the literal meaning and that of the strong meaning, but not that of the weak meaning. Overall, these results suggest that the weak reading may not be accessed in comprehension, at least in the absence of a context that could make the weak reading specifically relevant. This seems to favor approaches that do not predict a weak reading.

Two additional observations motivated two subsequent experiments, presented in Section 5. First, a natural and plausible hypothesis is that gradience within the

Predictor	Estimate	Standard error	<i>t</i> -value
(Intercept)	2.561	1.049	2.441
c_{vrf}	3.180	0.256	12.413
c_{lit}	69.076	0.573	120.579
c_{str}	10.197	1.109	9.192

Table 1: Fixed effects of the best-fitting linear mixed-effects model (bare plurals experiment).

literally true conditions can be explained by their proximity to the uniformly-plural scenario. In that case, one should still observe gradience in the interpretation of a sentence like “Each box contains [several NPs]”, which has no ambiguity and is interpreted as equivalent to the strong reading. Section 5.1 presents a comprehension study on *several NPs*.

Secondly, we were surprised by the high mean score (over 70%) received by the uniformly-singular scenario. It is conceivable that this very high rating produced a kind of ceiling effect: the space that remains between the literal and the strong conditions (from 72% to 95%) might be too narrow to detect the influence of all relevant factors, including the influence of the weak reading, in the presence of gradient effects. This unexpectedly high rating of the literal condition led us to investigate whether participants might have relied on a marginal cumulative interpretation, i.e. a reading whereby the sentence is judged true as long as there is, *in total*, a plurality of geometric shapes contained across all boxes, even when each box contains only one shape. If so, it seemed plausible to us that such a cumulative interpretation, which in principle should be at best marginal with a distributive quantifier such as “each”, would be less accessible with *some NPs* than with bare plurals. If so, it might be worth replicating the previous experiment with *some NPs* instead of bare plurals, so as to address the possibility of a ceiling effect in our previous experiment. Section 5.2 presents a comprehension study on the cumulative interpretation of three plural expressions under “each”: bare plurals, *some NPs* and *several NPs*.

5 Follow-up experiments

We present here two follow-up experiments aimed at shedding light on the source of gradient effects (Section 5.1) and on the availability of cumulative readings (Section 5.2), as we just discussed.

Rank	Model formula	BIC	AIC
1	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	44940	44901
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	44948	44902
3	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + (1 \mid \text{participant})$	45015	44980
4	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	45019	44983
5	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	45058	45018
6	$\text{response} \sim c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	45083	45051
7	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	45353	45320
8	$\text{response} \sim c_{\text{lit}} + (1 \mid \text{participant})$	45449	45423
9	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	48255	48216
10	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	48500	48467
11	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	48600	48568
12	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	48644	48617
13	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	51677	51644
14	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	51707	51680
15	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	52088	52062
16	$\text{response} \sim 1 + (1 \mid \text{participant})$	52556	52537

Table 2: Model comparison by BIC and AIC (bare plurals experiment).

5.1 Comprehension study on *several NPs*¹⁰

With *several NPs*, there are no levels of readings involved, as multiplicity is encoded in the quantifier’s denotation. As only one reading can be true, this is helpful for testing a hypothesis related to the source of gradient effects: they might reflect proximity to the closest situation that makes a certain reading true. Thus, we focused on observing whether gradience could still be found across conditions in which “Each box contains [several NPs]” is false, that is, 8 conditions out of 9. One reason why we might expect such gradient effects to arise with *several NPs* is that we observed gradient effects within the **FALSE** scenarios in the bare plurals experiment.

5.1.1 Methods

Methods were mostly identical to those of the bare plurals experiment (Section 4), with the following differences:

¹⁰ All preregistration materials for our experiments are available anonymously in [this OSF project](#). A link to the original pre-data-collection registration will be added after review.

- (i) **Participants.** We tested 70 adult participants recruited through Prolific (mean age 37.1; age range 19-72; 39 female). Participants were paid £0.60 for their participation. They had to meet the same eligibility criteria as in the previous experiment and must not have participated in the previous experiment. The sample size was based on power estimates using the data collected through the bare plurals experiment and was preregistered.
- (ii) **Stimuli.** Sentences in the trials followed the fixed structure: “Each box contains [several NPs]”. The rest of the design remained the same as before.
- (iii) **Experimental conditions.** These were identical to the bare plurals experiment, only the labeling of the conditions was different. As before, a box containing multiple shapes was called a *strong verifier*. Conditions were labeled based on whether they make the sentence true or false (**FALSE**, **STRONG**), followed by the number of empty boxes, then by the number of strong verifiers in the picture (see Figure 8).

5.1.2 Results

As in the bare plurals experiment, participants who completed the full experiment were excluded if:

- (i) they failed the attention check (0 participants);
- (ii) they did not rate every trial of the condition **STRONG-0-4** higher than every trial of the condition **FALSE-4-0** (0 participants).

Results are shown in Figure 9.

In this experiment, we only have one predictor, that is c_{vrf} , the number of strong verifiers. We subset the no-empty-box conditions (**FALSE-0-0**, **FALSE-0-1**, **FALSE-0-2**, **FALSE-0-3**, **STRONG-0-4**) and fit a linear mixed-effects model predicting responses as a function of c_{vrf} , with random intercepts and slopes by participant.

$$\text{response} \sim c_{\text{vrf}} + (1 + c_{\text{vrf}} \mid \text{participant})$$

We compared this model to a null model containing only the intercept, through LRT: the model fit significantly outperformed the null model ($\chi^2(1) = 232.31$, $p < 10^{-49}$). We did the same for the subset of at-least-one-empty-box conditions (**FALSE-4-0**, **FALSE-3-1**, **FALSE-2-2**, **FALSE-1-3**), with the same conclusion ($\chi^2(1) = 380.01$, $p < 10^{-81}$).¹¹ There were no convergence issues or singular fit for either model. Tables 3 and 4 report statistical details of the two fitted models.

¹¹ The p -values reported have been corrected for multiple comparisons using the Holm-Bonferroni method.

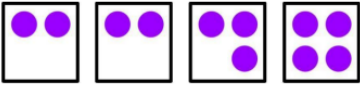
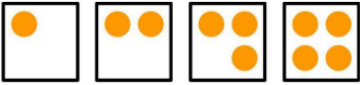
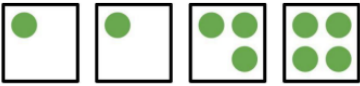
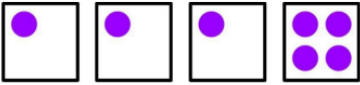
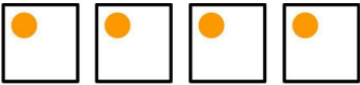
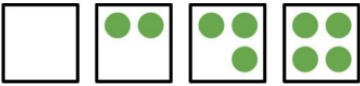
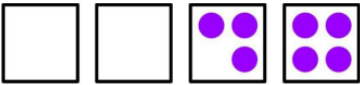
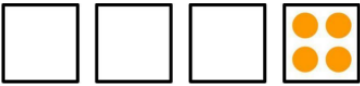
stimuli pictures (examples with circles)	label for condition
	STRONG-0-4
	FALSE-0-3
	FALSE-0-2
	FALSE-0-1
	FALSE-0-0
	FALSE-1-3
	FALSE-2-2
	FALSE-3-1
	FALSE-4-0

Figure 8: Stimuli for *several NPs*.

Predictor	Estimate	Standard error	<i>t</i> -value
(Intercept)	16.321	2.471	6.604
c_{vrf}	5.498	0.781	7.036

Table 3: Fixed effects for the no-empty-box model.

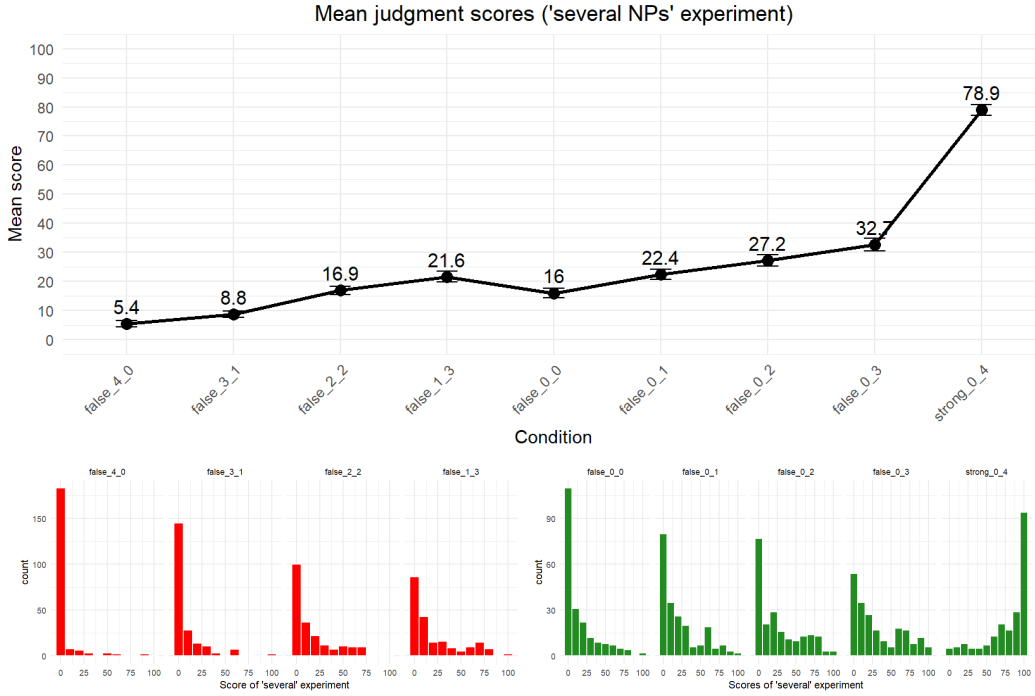


Figure 9: Results of the comprehension task with *several NPs*.

5.1.3 Discussion

We observed gradient effects driven by c_{vrf} within both subsets of conditions, which can be plausibly interpreted as reflecting increasing proximity to situations making the sentence true.¹² This supports the view that the gradient effects we observed in the previous experiment with bare plurals itself reflects proximity to the closest situation making the strong reading true.

¹² The mean rating is not only increasing with the number of strong verifiers, it is also higher on average when there is no empty box. This is expected given that a weak verifier (a box with just one item) is intuitively closer to a strong verifier than an empty box is.

Predictor	Estimate	Standard error	t-value
(Intercept)	4.677	1.176	3.978
c_{vrf}	5.653	0.966	5.852

Table 4: Fixed effects for the at-least-one-empty-box model.

A somewhat surprising observation is that the mean rating for **STRONG-0-4** is not extremely high, at 78.9. In comparison, the same condition received a mean rating above 90 in the bare plurals experiment,¹³ possibly reflecting the common interpretation of “several” as denoting a number greater than or equal to 3, rather than greater or equal to 2.

5.2 Comprehension study: cumulativeness of different plural expressions¹⁴

We now turn to a second follow-up experiment in which we investigate the availability of cumulative interpretations for different plural NPs under the scope of “each”.

5.2.1 Methods

Participants. Based on pilot data, and following our preregistered sample size, we recruited 100 participants per condition, hence 600 adults in total through Prolific, (mean age 42.6; age range 18-80; 328 female). They were paid £0.15 for their participation. We used the same eligibility criteria as in the previous experiments and excluded participants who had participated in the previous experiments.

Procedure and stimuli. A trial consisted of a picture paired with a sentence. The sentence followed a fixed structure: “[Each/The] box contains [plural expression]”. The image consisted of either one box or four boxes, each containing exactly one circle.

The experiment crossed two factors, yielding six conditions:

- **F1. Plural expression:** *several NPs*, *some NPs*, or bare plural;
- **F2. Picture type:** either a single box (sentence began with “The box”), or four boxes (sentence began with “Each box”).

Each participant completed exactly one trial. Assignment to condition was randomized, resulting in approximately 100 participants per condition. The rest of the design was identical to the previous experiment. Figure 10 shows two trials.

¹³ And also above 90 in the *some NPs* experiment, reported in Section 6

¹⁴ All preregistration materials for our experiments are available anonymously in [this OSF project](#). A link to the original pre-data-collection registration will be added after review.

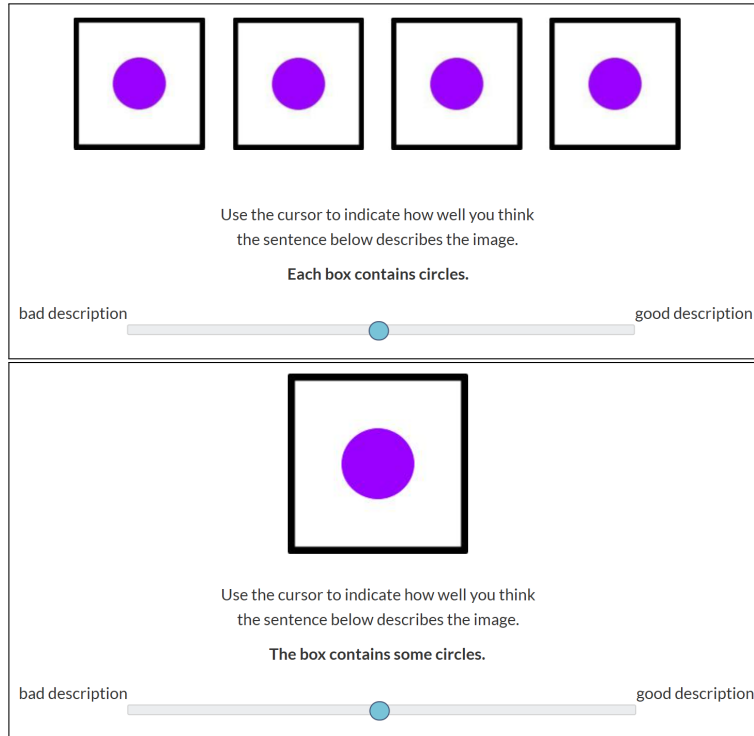


Figure 10: Example of trials in the cumulativity experiment.

5.2.2 Results

Participants who did not complete the full experiment were excluded from the analyses.

The mean ratings for each condition are reported in Table 5. Across all sentence types, participants gave higher ratings to sentences paired with 4-box pictures than to those paired with 1-box pictures.

	Bare plural	<i>Some NPs</i>	<i>Several NPs</i>
1 box	32.8	18.9	3.4
4 boxes	75.8	46.8	14.5

Table 5: Mean ratings (scale 1-100) by sentence type and image type.

We examined the effect of picture type within each sentence type (Table 6). For each sentence type, sentences paired with 4-box pictures received higher acceptabil-

ity ratings than those paired with 1-box pictures. Positive estimates indicate higher ratings in the 4-box condition. All contrasts were evaluated using Wald t -tests.

Sentence type	Estimate	SE	df	t -value	p -value
BP	43.0	4.20	594	10.254	< 0.001
<i>several</i>	11.0	4.42	594	2.500	< 0.05
<i>some</i>	27.8	4.19	594	6.641	< 0.001

Table 6: Within-sentence contrasts between the 1-box and 4-box conditions.

To explore any interaction between sentence type and picture type, we conducted three pairwise interaction tests and found that the interaction between sentence type and picture type was also significant (Table 7). Interaction contrasts quantify differences in the size of the picture-type effect across sentence types. Positive estimates indicate that the picture-type contrast (4-box vs. 1-box) is larger for the first sentence type than for the second. Each contrast was evaluated using a Wald t -test based on estimated marginal means obtained via the `emmeans` package in R.¹⁵

Sentence contrast	Estimate	SE	df	t -value	p -value
BP vs. <i>several</i>	31.997	6.095	594	5.250	< 0.001
BP vs. <i>some</i>	15.203	5.933	594	2.563	< 0.01
<i>some</i> vs. <i>several</i>	16.795	6.091	594	2.757	< 0.01

Table 7: Pairwise interaction contrasts of picture type between sentence types. Positive estimates indicate larger picture-type effects for the first sentence type listed in the contrast.

5.2.3 Discussion

Our results supported the hypothesis that participants are sensitive to a possible cumulative interpretation when multiple boxes are shown in total.

The effect of *some NPs* across picture types was smaller in magnitude than the effect of bare plurals. Furthermore, the mean rating of the 1-box scenario was significantly lower with *some NPs* than with bare plurals. This is unaccounted for, as

¹⁵ All p -values reported have been corrected for multiple comparisons using the Holm-Bonferroni method.

existing theories do not predict different availability levels of cumulative plural readings between bare plurals and *some NPs*. The results also showed that *several NPs* resists cumulative interpretations.

6 Comprehension study on *some NPs*: continuous judgments¹⁶

As far as the theories presented in Section 3 are concerned, the predictions for bare plurals also apply to *some NPs*. However, as the cumulativeness experiment showed, the mean rating for the uniformly-singular condition is 20 points lower with *some NPs* than with bare plurals. This suggests that there is more room for *some NPs* ratings to increase within literally true conditions. It is possible that, in the bare plurals experiment, the combination of high baseline ratings (i.e. ratings in the uniformly-singular scenario) and gradient within literally true conditions masked the qualitative shift from the literal to the weak reading. If so, *some NPs* should provide better conditions for observing this shift.

6.1 Methods

Methods were mostly identical to those of the comprehension study on bare plurals (Section 4), with the following differences:

- (i) **Participants.** After exclusions, we tested 200 adult participants recruited through Prolific (mean age 41.2; age range 18-73; 88 female). Participants were paid £0.60 for their participation. They had to meet the same eligibility criteria as in the previous experiments and must not have participated in the previous experiments.
The sample size was based on power estimates using the data collected through the bare plurals experiment.
- (ii) **Stimuli.** Sentences in the trials followed the fixed structure: “Each box contains [some NPs]”. The rest of the design remained the same as before.

6.2 Results

Results are shown in Figure 11.

We preregistered the same analyses as for the bare plurals experiment and included the model comparison using BIC. We fit a mixed-effects model to the subset of literally true conditions, with c_{vrf} as the sole predictor. The random slope was removed due to failure of convergence. This significantly outperformed the null model

¹⁶ All preregistration materials for our experiments are available anonymously in [this OSF project](#). A link to the original pre-data-collection registration will be added after review.

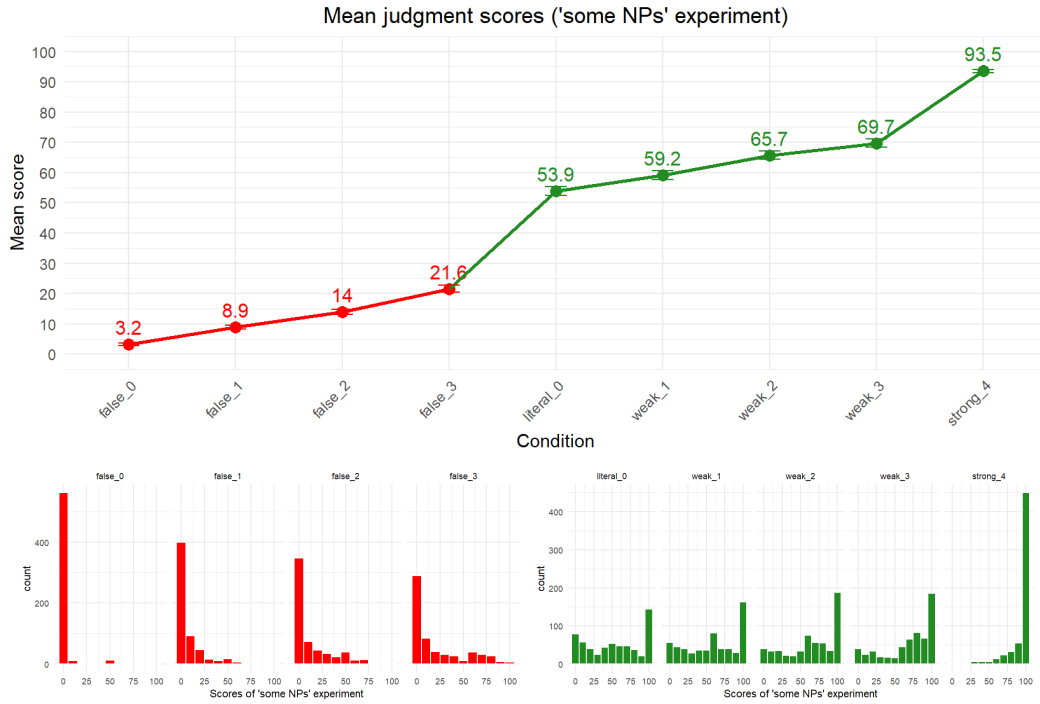


Figure 11: Results of the comprehension task with *some NPs* (continuous judgments).

($\chi^2(1) = 1052.9, p < 10^{-15}$).¹⁷ As expected, the effect of c_{vrf} was positive ($b = 8.98$, $SE = 0.25$, $t = 35.75$).

A gradient effect was also found within **WEAK** conditions alone ($\chi^2(1) = 65.19$, $p < 10^{-15}$), where the random slope allowed model convergence and was kept.

We then compared all mixed-effects linear models based on all possible combinations of predictor, with a random intercept per participant. The best-fitting model across all 9 conditions was the same as for the bare plurals experiment:

$$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$$

Table 8 reports statistical details of the best model.

The second best-fitting model was, again:

$$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$$

with $\Delta BIC = 9$ between the two best models. This difference is considered strong evidence in favor of the better-ranked model.

¹⁷ The p -values reported have been corrected for multiple comparisons using the Holm-Bonferroni method.

Predictor	Estimate	Standard error	<i>t</i> -value
(Intercept)	3.361	1.284	2.618
c_{vrf}	5.716	0.272	20.989
c_{lit}	50.190	0.609	82.419
c_{str}	17.108	1.179	14.508

Table 8: Fixed effects of the best-fitting linear mixed-effects model (*some NPs* experiment, continuous judgments).

The full model ranking using BIC is presented in Table 9 (for AIC results, see Table 13 in Appendix A). The order of ranking using BIC is identical to the one obtained from the bare plurals data (Table 2). While the order differs between AIC and BIC (only by the relative positions of the models of rank 3 and 4), both criteria agree on the two best models.

6.3 Discussion

Consistent with the findings from the cumulativity experiment (Section 5.2), baseline judgments for the **LITERAL-0** condition were lower with *some NPs* than with bare plurals. We still did not observe any qualitative difference between the **LITERAL-0** and **WEAK** conditions. As in the bare plurals experiment, the model comparison using BIC ranked the model without c_{weak} , but with all three other predictors, as the best. Model comparison therefore goes in favor of theories that generate the literal and strong readings but not the weak reading.

7 Comprehension study on *some NPs*: binary judgments¹⁸

We conducted a second version of the experiment using binary truth-value judgments, in order to see whether gradient effects would still arise with binary judgments and whether the same readings would be accessed in the best model.

7.1 Methods

Methods were mostly identical to those of the comprehension study on *some NPs* with continuous judgments (Section 6), with the following differences:

¹⁸ All preregistration materials for our experiments are available anonymously in [this OSF project](#). A link to the original pre-data-collection registration will be added after review.

Rank	Model formula	BIC
1	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	48869
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	48878
3	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + (1 \mid \text{participant})$	49067
4	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	49070
5	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	49178
6	$\text{response} \sim c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	49283
7	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	49899
8	$\text{response} \sim c_{\text{lit}} + (1 \mid \text{participant})$	50173
9	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	50706
10	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	50707
11	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	51178
12	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	51250
13	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	53206
14	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	53385
15	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	53761
16	$\text{response} \sim 1 + (1 \mid \text{participant})$	54771

Table 9: Model comparison by BIC (*some NPs* experiment, continuous judgments).

- (i) **Participants.** After exclusions, we tested 200 adult participants recruited through Prolific (mean age 42.0; age range 18-76; 93 female). They had to meet the same eligibility criteria as in the previous experiments and must not have participated in the previous experiments.
- (ii) **Stimuli.** The instructions accompanying each trial were: “Do you think the sentence below is true or false?”, with the options “false” on the left and “true” on the right.

7.2 Analyses

As the response type was binary, we analyzed the data using logistic mixed-effect models, and not linear mixed-effect models as we did with the continuous judgments. The rest of this analysis was identical to that conducted for *some NPs* with continuous judgments, with random intercepts by participant as random effects.

Results are shown in Figure 12. There is hardly any observable gradience within one same level of reading (**FALSE** or **WEAK**).

While fitting a mixed-effects model to the subset of literally true conditions, the random slope was removed due to failure of convergence. The model fit on lit-

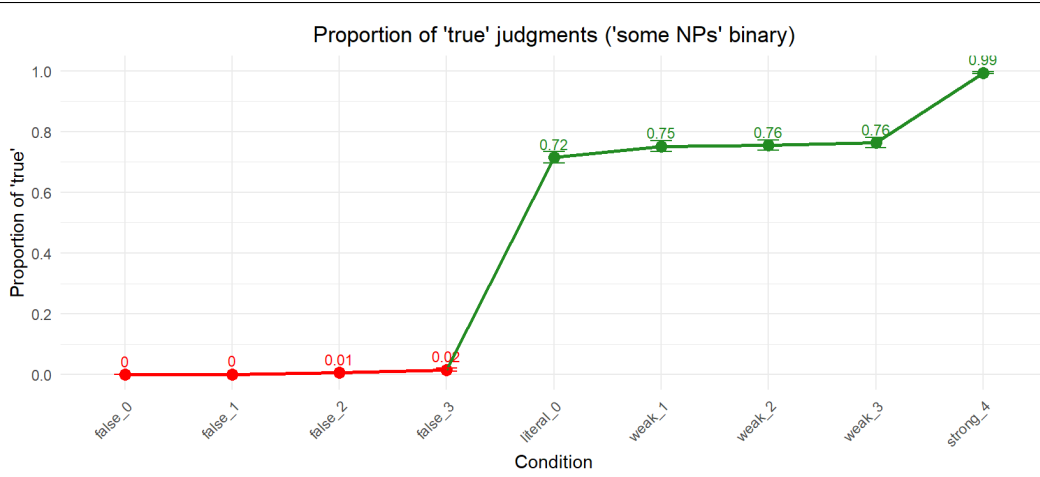


Figure 12: Results of the comprehension task with *some NPs* (binary judgments).

erally true conditions (c_{vrf} as predictor) significantly outperformed the null model ($\chi^2(1) = 400.28, p < 10^{-15}$)¹⁹. As expected, the effect of c_{vrf} was positive ($b = 1.54, SE = 0.11, t = 14.03$).

However, within **WEAK** conditions alone, the LRT yielded $\chi^2(1) = 1.77$ and $p = 0.183$, suggesting that adding c_{vrf} did not significantly improve model fit compared to the null model.

The best-fitting model across all 9 conditions was the same as for the previous experiments:

$$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$$

The second best-fitting model was, again:

$$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$$

with $\Delta BIC = 8$ between the two best models. This difference is considered strong evidence in favor of the better-ranked model.

The full model rankings using BIC is presented in Table 10 (for AIC results, see Table 14 in Appendix A). The order of ranking using BIC is not fully identical to either the one obtained through AIC, or the one obtained from the data of *some NPs* with continuous judgments (Table 9). However, the differences in ranking are not concerned with the two best models.

¹⁹ The p -values reported have been corrected for multiple comparisons using the Holm-Bonferroni method.

Rank	Model formula	BIC
1	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	1218
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	1226
3	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	1227
4	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + (1 \mid \text{participant})$	1240
5	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	1433
6	$\text{response} \sim c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	1452
7	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	1749
8	$\text{response} \sim c_{\text{lit}} + (1 \mid \text{participant})$	1820
9	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	2959
10	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	3500
11	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	3819
12	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	3921
13	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	5972
14	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	5978
15	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	6771
16	$\text{response} \sim 1 + (1 \mid \text{participant})$	7149

Table 10: Model comparison by BIC (*some NPs* experiment, binary judgments).

7.3 Discussion

Visually, three levels can be distinguished: the **FALSE** level, the **LITERAL+WEAK** level, and the **STRONG** level. Contrary to our experiments with continuous judgments, gradience seems to be nearly non-existent here within the first two levels. However, model comparisons with BIC as well as AIC still select a model with the c_{vrf} factor as the best model.

Before taking this result at face value, we must be cautious about the properties of logistic regression. Indeed, the logistic model has limitations for probabilities near 0 or 1. In the **FALSE** conditions, the mean scores are very close to 0 and tiny differences in scores will be amplified in terms of log-odds. Because the log-odds difference between 0.01 and 0.02 is huge, the tiny ‘gradient’ effect in **FALSE** cases (which is theoretically irrelevant) could misleadingly drive the model to favor inclusion of the gradience predictor c_{vrf} . The equally tiny gradient effect within **WEAK** conditions was shown to be non-significant by an LRT test.

As a sanity check, and not as part of a preregistered analysis, we conducted model comparisons only with the data from the literally true conditions. The predictor c_{lit}

was therefore removed from the combinations of predictors. We expected the best model to include c_{str} , as the graph showed a stark increase in ratings in the uniformly-plural scenario. The model ranking using BIC is presented in Table 11 (for AIC results, see Table 15 in Appendix A).

Rank	Model formula	BIC
1	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	937
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	941
3	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	944
4	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	954
5	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	1170
6	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	1187
7	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	1497
8	$\text{response} \sim 1 + (1 \mid \text{participant})$	1579

Table 11: Model comparison by BIC (*some NPs* experiment, binary judgments, subset of literally true conditions).

The best-fitting model across the subset of 5 conditions was

$$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$$

Table 12 reports statistical details of the best model.

Predictor	Estimate	Standard error	z-value
(Intercept)	8.606	0.664	12.961***
c_{weak}	1.394	0.289	4.816***
c_{str}	9.426	0.738	12.770***

Table 12: Fixed effects of the best-fitting logistic mixed-effects model (*some NPs* experiment, binary judgments, literally true conditions).

Among the various experimental designs we have presented, the weak reading emerged in the best model only when the response task was binary and when the analysis was restricted to the subset of literally true conditions. On the one hand, this finding could have important theoretical implications, as it may constitute evidence that the weak reading is indeed accessible, contrary to what the best-fitting models

from all continuous response tasks suggest.²⁰ On the other hand, only a post-hoc analysis detects the weak reading, and the effect is due to a very small proportion of participants.²¹ Even if the weak reading might be available, overall it seems to be much less salient than the literal and strong readings.

8 Clustering of individual response profiles

Clustering can be useful to detect tendencies in participants' behaviors that are not clearly visible when only looking at aggregated scores. In particular, it allows us to see whether participants behave in a relatively uniform manner, with intermediate judgments reflecting genuinely intermediate responses at the individual level, or whether these intermediate patterns arise from averaging over participants who in fact adopt different, more categorical interpretations. In other words, do speakers consistently access a single reading, or might different groups of speakers favor different readings? Clustering helps identify whether participants form groups with similar response patterns. We begin by visualizing individual response profiles superimposed in the same graph, where each line represents a participant's mean judgments in each condition (Figure 13). These plots already show substantial variability: some participants display almost binary patterns, though at different thresholds, while others show more gradient or more irregular trajectories.

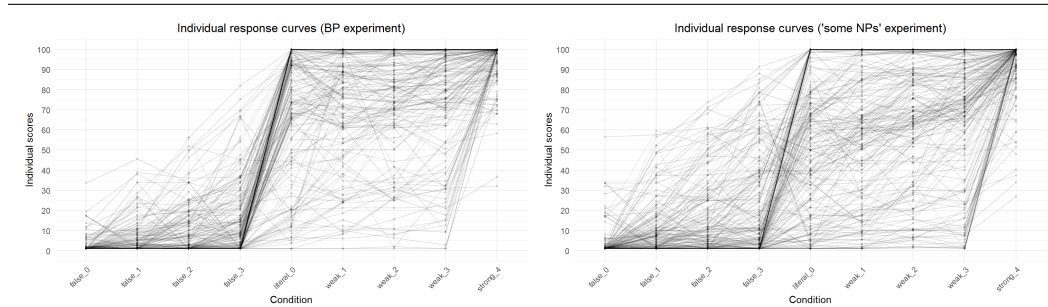


Figure 13: Individual response profiles in the BP (left) and the continuous *some NPs* (right) experiments.

20 We conducted a similar analysis over the data from the previous experiment (continuous judgments with *some NPs*). In this case, the best model included c_{vrf} and c_{str} as fixed predictors, but not c_{weak} . The second best model included c_{weak} , with $\Delta BIC = 8$. This additional analysis does not suggest that the weak reading was accessed in this experiment.

21 Out of 200 participants, 175 endorsed as true the **WEAK-1** and **LITERAL-0** to the same extent, 18 preferred **WEAK-1** and 7 preferred **LITERAL-0**. In contrast, the difference between the **STRONG** and the **WEAK-3** conditions is much more pronounced, with 53 participants preferring **STRONG** over **WEAK-3** vs. 3 showing a reverse preference.

We then group participants based on the similarity of their full response profiles. Intuitively, the goal is to partition participants into clusters such that individuals within a cluster behave as similarly as possible, while individuals in different clusters behave as differently as possible. Concretely, each participant is represented by a vector of mean ratings across conditions, and participants are grouped according to how close these vectors are to one another.²²

We applied this same procedure to three datasets: BP, continuous *some NPs*, and binary *some NPs*. The results are shown in Figure 14. In all three cases, the optimal solution partitions participants into two clusters. This supports theories that predict two readings. These two clusters can be loosely described as corresponding to a group that tends to access the literal reading and a group that tends to access the strong reading.

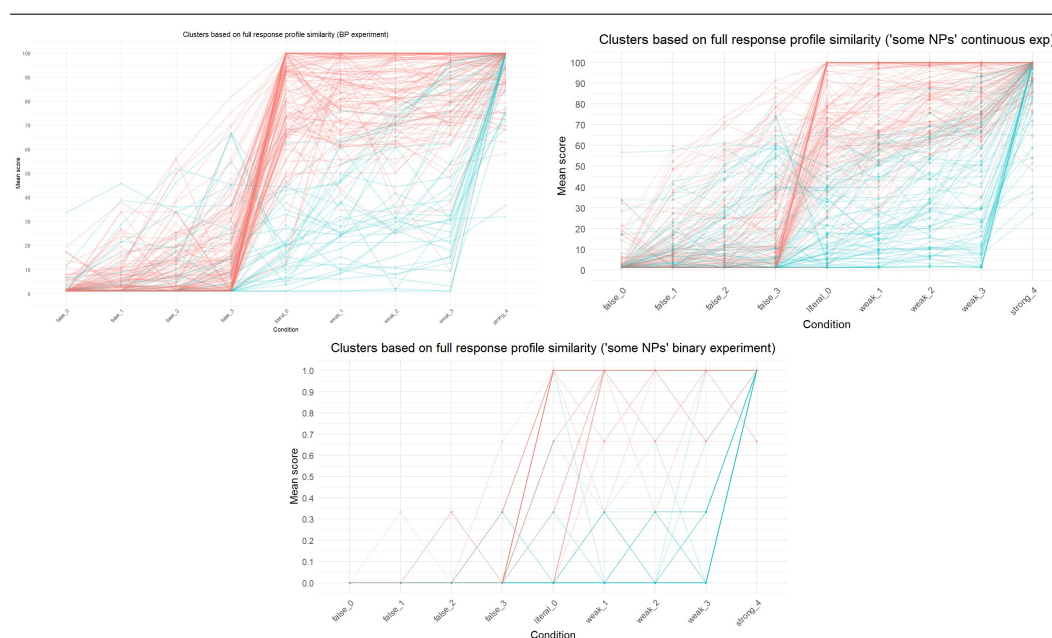


Figure 14: Clustering over full profiles in the BP (top left) and *some NPs* experiments (continuous response: top right; binary response: bottom).

Importantly, this interpretation should be treated with caution. Firstly, the clusters are not internally homogeneous: even within the same cluster, participants still

²² We implement this in R using hierarchical clustering (via `hclust`) over Euclidean distances between response profiles (computed with `dist`). Cluster assignments are obtained by cutting the hierarchical tree into k groups, using `cutree`. The number of clusters is chosen by comparing different values of k and selecting the one that maximizes the average silhouette score (computed with `silhouette`), which measures how well each participant fits within its assigned cluster compared to other clusters.

show quite a lot of variation in the shape of their response profiles. Secondly, the plots in Figure 13 suggest the presence of intermediate patterns that are not cleanly captured by a simple binary partition. While the clustering results concur with our other results in supporting theories that predict two readings, we still take them only as indicating broad tendencies in the data.

9 Conclusion

Our inquiries were guided by two questions: the theoretical question of which readings are available, and the methodological question of how we can experimentally disentangle readings from gradient effects.

On the theoretical side, model comparisons in experiments involving continuous judgments suggest that there is strong evidence for the availability of the literal and strong readings, as well as for gradient effects, but the best models do not include the weak reading.²³ At the same time, post-hoc analyses find evidence for the weak reading in the one experiment that involved binary data. One might therefore wonder (a) whether gradient effects were not in fact a task effect, and (b) whether such gradient effects did not ‘mask’ the weak reading. While this is a possibility, it is important to note that this apparent discrepancy can also be explained quite differently. If underlyingly, judgments are in fact gradient, one should view binary answers as resulting from individuals dichotomizing their underlyingly continuous scale; that is, individuals would answer with ‘true’, in a binary task, when the underlying continuous judgment is above a certain threshold. For some distributions of thresholds across participants, a ‘leap’ between two conditions could appear in binary judgments even if there is no such leap in the underlying continuous judgments. Even if clustering analyses suggest that ‘typical’ participants tend not to use intermediate values, there are still a lot of participants who use the full extent of the scale. In any case, while we cannot exclude the possibility that the weak reading is available, our results overall, including the binary task results, suggest that this reading, if it exists, is much less salient than the literal or strong reading. Even in the binary task, we detected the weak reading only in a post-hoc analysis (and not in the preregistered analysis), and the effect is quantitatively small.

We did find clear gradient effects in the continuous judgment tasks. While in principle such gradient effects might reflect a notion of ‘typicality’ that is independent of truth-conditions (as suggested in van Tiel & Geurts 2014, van Tiel 2014), our data, especially in light of our follow-up experiment based on *several*-sentences, suggest that gradience reflects something like similarity to a situation that makes the strong reading true. That is, among situations that make the strong reading false,

²³ Revisiting the data from Chemla & Spector (2011), we arrived at the same conclusions, although their experiments were aimed at detecting local implicatures and not the weak reading.

those that are closer to a case making the strong reading true are rated higher (cf. Chemla & Spector 2011 and van Tiel et al. 2021). Of course, in principle, we should expect all available readings to contribute to gradient effects. In our design, however, these cumulative gradient effects of different readings are confounded: in the **FALSE** conditions, as the number of strong verifiers increases, the image becomes closer to situations that make true the literal reading, the weak reading, and the strong reading; in the other conditions, the literal reading is always true, which can generate no gradient effects; the weak reading is only false in the **LITERAL** condition, hence cannot lead to observable gradient effects.

Our findings seem to favor theories that predict two readings, the literal one and the strong one (i.e. Zweig 2009 and trivalent approaches such as Ahn et al. 2020 and Križ 2016, or a modified, constrained version of the higher-order implicature approach proposed in Spector 2007, as well as the ambiguity approach of Farkas & de Swart 2010). However, there are several such theories and our design does not adjudicate between them. Future work could test, for instance, cases where bare plurals are embedded under plural non-monotonic quantifiers (e.g. “exactly three”), where different theories make subtly different predictions.

Our results also have significant methodological implications. In particular, in non-binary judgment tasks (be they continuous or based on a discrete scale), inferences about readings cannot be safely reached without controlling for gradient effects.

Competing interests

The authors have no competing interests to declare.

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A AIC tables

Rank	Model formula	AIC
1	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	48830
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	48831
3	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	49031
4	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + (1 \mid \text{participant})$	49034
5	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	49139
6	$\text{response} \sim c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	49251
7	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	49866
8	$\text{response} \sim c_{\text{lit}} + (1 \mid \text{participant})$	50147
9	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	50666
10	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	50674
11	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	51145
12	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	51224
13	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	53173
14	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	53359
15	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	53734
16	$\text{response} \sim 1 + (1 \mid \text{participant})$	54751

Table 13: Model comparison by AIC (*some NPs* experiment, continuous judgments).

Rank	Model formula	AIC
1	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	1185
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	1186
3	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	1194
4	$\text{response} \sim c_{\text{lit}} + c_{\text{str}} + (1 \mid \text{participant})$	1213
5	$\text{response} \sim c_{\text{vrf}} + c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	1400
6	$\text{response} \sim c_{\text{lit}} + c_{\text{vrf}} + (1 \mid \text{participant})$	1426
7	$\text{response} \sim c_{\text{lit}} + c_{\text{weak}} + (1 \mid \text{participant})$	1723
8	$\text{response} \sim c_{\text{lit}} + (1 \mid \text{participant})$	1800
9	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	2926
10	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	3474
11	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	3792
12	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	3901
13	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	5951
14	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	5952
15	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	6752
16	$\text{response} \sim 1 + (1 \mid \text{participant})$	7136

Table 14: Model comparison by AIC (*some NPs* experiment, binary judgments).

Rank	Model formula	AIC
1	$\text{response} \sim c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	913
2	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + c_{\text{str}} + (1 \mid \text{participant})$	914
3	$\text{response} \sim c_{\text{vrf}} + c_{\text{str}} + (1 \mid \text{participant})$	917
4	$\text{response} \sim c_{\text{str}} + (1 \mid \text{participant})$	936
5	$\text{response} \sim c_{\text{vrf}} + c_{\text{weak}} + (1 \mid \text{participant})$	1146
6	$\text{response} \sim c_{\text{vrf}} + (1 \mid \text{participant})$	1169
7	$\text{response} \sim c_{\text{weak}} + (1 \mid \text{participant})$	1479
8	$\text{response} \sim 1 + (1 \mid \text{participant})$	1567

Table 15: Model comparison by AIC (*some NPs* experiment, binary judgments, subset of literally true conditions).